

SCOUT → Master Planner: Handoff Report (Basketball / Sports Domain)

Agent: SCOUT (Cursor agent; NCAA basketball panel, Tier 1 modeling, generative simulation, HPC sweeps)

Domain root: `./sports/` (panel + Tier 1 modules; shared panel CSV at workspace `datasets/mbb/`)

Report date: 2026-06-08

Prepared for: Charles Levine → future **Master Planner** agent (Cursor)

Charles's explicit instruction (2026-06-08): This report feeds the Master Planner. **Do not** begin drafting the cross-project master plan until Charles **assigns** that agent and asks for planning work. SCOUT/PEER/CODA/VECTOR should only **report status** until then.

Companion reports (same folder): - `CODA_report_to_master_planner.md` — Army / `talent/` - `PEER_report_to_master_planner.md` — Academia / `tenure/` - **VECTOR report:** Not yet in repo as of this writing — expect manuscript-focused handoff from Scholar GPT workflow or a Cursor session dedicated to `5-Manuscript/`.

0. What this document is (and is not)

What it is

- A **ground-truth status snapshot** of the basketball/sports lane as SCOUT understands it after work through **June 2026** (530 inverted-U replicated; 538 empirical Wang ladder; 538D generative CELL 10 congestion score; axis-conditioning finding; documentation pass).
- A **file map** so the Master Planner can navigate without re-deriving context from chat logs, SpecStory exports, or stale May briefs.
- A **list of open questions** where Charles's intent is not fully locked — SCOUT prefers explicit answers over assumptions.
- Pointers to **CODA**, **PEER**, and **VECTOR** without claiming authority over Army pipelines, tenure scrape, or manuscript prose.

What it is not

- **Not** the master plan itself.
 - **Not** a claim that generative simulation **replicates** the empirical inverted-U on the same conditioning axis as 530 (`poolq_loo` / L_Q LOO).
 - **Not** a substitute for reading `sports/documents/SPORTS_DATA_GAMEPLAN.md`, `tier_1_roadmap.md`, or `538_Cell10_Generative_Manual.md` when implementing or debugging notebooks.
-

1. Thesis frame (why SCOUT exists in this repo)

Charles's dissertation tests **advancement under constrained distinction**: individuals compete in **local talent pools** while advancement is **globally scarce**. Mid-quality peer environments can produce the highest individual advancement rates; elite environments create **congestion / signal compression** that drives rates back down (inverted-U).

Setting	Agent	Repo home	Outcome	Pool-quality construct
1 — Army	CODA	<code>talent/</code>	Promotion to Major vs attrition	Senior-rater LOO pool quality; CIF / Cox
2 — Basketball	SCOUT	<code>sports/</code>	NBA draft (<code>Y_draft</code>)	<code>poolq_loo</code> (LOO teammate <code>perf</code> , usually PPM z)
3 — Academia	PEER	<code>tenure/</code>	Tenure vs pre-tenure exit	<code>poolq_loo_mean</code> (LOO dept assistant pubs)

SCOUT’s job: Build and maintain the college basketball → NBA draft empirical stack (530 conductor, panel exports, forensics), the Alex-ordered Tier 1 modeling ladder (538 / 538D), and the modular generative lab (`tier1_pool_assignment.py`, CELL 10 playground) that stress-tests the minimal congestion score without over-claiming mechanism closure.

Why basketball is “Setting 2” in the cross-domain story: Global selection (draft) vs local opportunity (minutes, roles, teammate quality) is the cleanest measurement lab for Tier 1 objects (A, L_Q, L_C, Λ). Army is the anchor replication; academia is the third replication (preliminary per PEER report).

Cross-setting manuscript doc (VECTOR + Charles, June 2026):

`1-Various_PDE_and_Chat_stuff/5-Manuscript/advancement_under_constrained_distinction_dakota_feedback_v03.rtf` — committee-facing whole-project summary. Basketball is §1–§2 empirical anchor; §5–§7 generative framing aligns with SCOUT’s June status.

2. Current status — basketball / SCOUT (high level)

Established / working

Layer	Status	Primary artifacts / location
Panel conductor	✓	<code>sports/530_sports_pipeline.ipynb</code> → <code>datasets/mbb/player_season_panel_530.csv</code>
Empirical inverted-U	✓	Binned draft rate vs <code>poolq_loo</code> ; exports under <code>datasets/mbb/exports_inverted_u_v0/</code> (e.g. 2026-06-02 run)
Wang empirical ladder	✓	<code>sports/538_alex_tier1_model_and_fit.ipynb</code> — CELLS 4–6 (bins → LPM → logit, L^*)
Pool forensics (PD11 targets)	✓	<code>530</code> CELLS 5–9 — overlap, mean×SD, draftee-only variants
Tier 1 mechanism columns	✓	<code>sports_pipeline/tier1_mechanism_vars.py</code> — <code>congestion_quality</code> , crowding variants
Soft generative assignment	✓	<code>sports/tier1_pool_assignment.py</code> — $\tau \approx 0.65$ default, Plot A overlap calibration
Congestion selection score	✓	$S_i = A_i - w \cdot L_C$ with <code>crowding_smooth</code> ; <code>crowding_l_z_scale</code> for z-scored ability
Generative playground UI	✓	<code>sports/538D_development.ipynb</code> CELL 10; <code>tier1_cell10_playground_run.py</code>
539 score preset (partial DGP)	✓	Green button / <code>SELECTION_539_*</code> in <code>tier1_sim_config.py</code> — score + [0,1] scales, not full 539 assignment/noise/threshold
Plot B axis toggle	✓	<code>SHOW_PLOT_B_TEAM_MEAN</code> — <code>False</code> = L_Q LOO (530); <code>True</code> = team_mean (539-style)
HPC parameter sweeps	✓ (infrastructure)	<code>sports/outputs/simulation_sweeps/</code> — Rivanna/Mac faithful 538 sweep runbooks

Major scientific finding (June 2026 — Master Planner must not garble this)

Same congestion score, different Plot B x-axis → different curve shape.

Plot B conditioning	Typical shape with 539 selection preset	Interpretation
team_mean (<code>SHOW_PLOT_B_TEAM_MEAN=True</code>)	Inverted-U (rise, peak, right-tail dip)	Matches 539 notebook readout (success vs team quality)
L_Q LOO (<code>SHOW_PLOT_B_TEAM_MEAN=False</code>)	Mostly decreasing	Matches 530 empirical axis question — standouts in weak peer pools selected most

Honest claims for manuscript / briefings: -  Empirical inverted-U on **real** college rosters vs LOO `poolq_loo` — **replicated** (stylized fact, not mechanism proof). -  Minimal score **equation** implemented: advancement score ↑ in own ability, ↓ in local congestion. -  Generative **proof-of-concept**: inverted-U vs **team_mean** under congestion; **not** bin-for-bin match on L_Q with same top- K knobs. -  Do **not** say “538D decomposes 539” or “generative replicates 530 on the same axis without qualification.”

Authoritative narrative doc: `Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md`

Three-level Wang sequence (oral + manuscript spine)

Level	Content	SCOUT status
0 — Equation	$S_i = A_i - \lambda L_{C,i}$	Written + coded
1 — 539 bundled POC	Sort-chop, score noise ϵ , 90th percentile threshold, plot vs team_mean	<code>539_alex_model.ipynb</code> — reference DGP, not architecture parent
2 — 538D modular lab	Soft assign $\tau \approx 0.65$, top- K , no score noise on main path, plot axis swappable	Active development notebook

Not done / open (sports-specific)

Item	Notes
Generative match to empirical L_Q inverted-U	Open — may need assignment noise placement, threshold rule, development term, or different selection rule
538 empirical CELL 7+	Robustness (FE, clustering) — placeholder
535 roadmap TODOs	Cleaner CELL 5 regression block; theory language distinguishing <code>poolq_loo</code> vs mechanism columns
CELL 4D heterogeneity	Wired in 538D (<code>HETEROGENEITY_TOP_TAIL</code>); narrative parked until scaffolding clearer
537 legacy sim	Frozen — sort-and-chop; use only for Thread C promotion-score experiments or benchmark overlay
SR / BPM coverage gaps	Small colleges, 404 slugs — documented in <code>Pertinent_Thoughts_Scout.md</code>
Draft matcher 2c	Human label loop in gameplan still “planned”
Formal master plan	Deferred until Charles assigns Master Planner

3. Notebook and module architecture (SCOUT conductor map)

530_sports_pipeline.ipynb - DATA: spine → draft match → panel → ventile EDA / LPM
535_sports_tier_1.ipynb - MECHANICAL EDA: tier1_mechanism_vars, CELL 4 plots (roadmap contract)
538_alex_tier1_model_and_fit.ipynb - EMPIRICAL WANG LADDER + generative CELL 10–12 (Alex spine)
538D_development.ipynb - PRIMARY June 2026 generative lab (+ 4D heterogeneity)
537_Sports_Simulation.ipynb - FROZEN legacy (sort-and-chop, old Cell 10)
539_alex_model.ipynb - Bundled generative POC (one-night); reference only for score preset

File	Role
<code>sports/tier1_pool_assignment.py</code>	<code>soft_assign</code> , selection, <code>crowding_smooth</code> , scale helpers
<code>sports/tier1_sim_config.py</code>	Default constants, <code>SELECTION_539_*</code> , <code>SHOW_PLOT_B_TEAM_MEAN</code>
<code>sports/tier1_cell10_playground_run.py</code>	ipywidgets UI
<code>sports/tier1_generative_eda.py</code>	Plot B bin tables + figures
<code>sports/tier1_cell12_generative_eda.py</code>	Static replay from <code>tier1_cell10_playground_state.json</code>
<code>sports/tier1_heterogeneity_ventiles.py</code>	CELL 4D exports
<code>sports/tier1_539_reference_settings.json</code>	Reference λ , θ , γ for 539 preset

Package layer (`sports_pipeline/` — empirical panel)

Module	Role
<code>panel_build.py</code>	Panel construction, LOO recompute
<code>tier1_mechanism_vars.py</code>	<code>congestion_quality</code> , crowding columns
<code>paths.py</code>	<code>panel_530_csv()</code> → workspace <code>datasets/mbb/</code>

Workspace root panel path: `datasets/mbb/player_season_panel_530.csv` (not under `sports/datasets/`).

4. Key design decisions (basketball-specific)

- Realized pools vs generative rules are separate layers — 538 §6 measures pools the world formed; CELL 10 simulates assignment rules calibrated to 530 forensics (`Tier1_Presorting_Design_Note.md`).
- 539 vs 538D is parallel, not parent/child — 539 is a bundled proof-of-concept DGP; 538D is a modular lab anchored to 530 empirics. The minimal claim is the score equation, not either notebook.
- Plot axis is part of the estimand — Conditioning success on `poolq_loo` (LOO peer quality) vs `team_mean` is not interchangeable; June 2026 finding is a **feature** for distinguishing mechanism tests.
- τ calibration — Assignment temperature $\tau \approx 0.65$ targets 530 CELL 8 overlap (peak $\gg 1$, median roster SD ~ 0.8 z). Engineering detail for SCOUT; not centerpiece for VECTOR oral brief.
- Ability scale + congestion scale — When ability is z-scored, `crowding_1_z_scale` (p90–p10 of A) prevents crowding penalty from being negligible in selection.

6. **537 frozen** — Do not extend sort-and-chop for Thread A; red dashed benchmark overlay in Plot A only.
7. **Document-first split** — `SPORTS_DATA_GAMEPLAN.md` = strategic intent; `tier_1_roadmap.md` = 535 execution contract;
`Alex_Tier1_Sequential_Model_Outline.md` = advisor spine (5-Manuscript).
-

5. Documentation inventory (SCOUT-authored or SCOUT-maintained, updated June 2026)

sports/documents/ — primary SCOUT contracts

Document	Role	Last refreshed
SPORTS_DATA_GAMEPLAN.md	Forward-looking pipeline contract; fast-path  ; Stage 5 Tier 1/generative	2026-06-08
tier_1_roadmap.md	535  contract, CELL 0, 538/538D generative block	2026-06-08
Tier1_Presorting_Design_Note.md	PD11 threads A/B/C; 530 calibration targets	2026-06-08
538_Cell10_Generative_Manual.md	CELL 10 operator manual	2026-06-08
Pertinent_Thoughts_Scout.md	Scratch discoveries + June generative section	2026-06-08
Alex_model_interpreted.md	PD12 interpretation; viable-peer signals table	2026-06-08
RIVANNA_RUNBOOK.md	HPC workflow	(ops; verify before large sweeps)
Mac_Faithful_538_Sweep_For_Dummies.md	Local sweep	
Rivanna_Faithful_538_Sweep_For_Dummies.md	Rivanna sweep	
537_Manual.md	Legacy sim only	
One_Page_Advisor_Brief_Template_College_Replication.md	Fill-in template (not auto-updated)	

Document	Role	Notes
Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md	Canonical June ground truth for generative + briefing	Lead doc for VECTOR
Scout_Modeling_Status_for_Vector_Barabasi_Briefing.md	May brief	Superseded banner → June doc
Alex_Tier1_Sequential_Model_Outline.md	Advisor-ordered spine §6A/ §7/§10	Updated 2026-06-08
Vector_to_Scout_Tier1_Modeling_Direction.md	VECTOR→SCOUT sync	June addendum in Final Note
alex_gates_briefing_structure_outline_v4.md	5-minute oral structure	§5 generative status June
important_stuff.md	Charles scratch anchor lines	
Tier1_Briefing_Outline.md	Equations + data map	Theory; not execution contract
Tier1_Narrative_Outline.md	Prose / competing forces	
tier_1_model.md	Model layer reference	

Cross-domain docs SCOUT reads but does not own

Document	Agent
talent/documents/Coda_Summary_For_Scout_and_Vector_Post_Replication.md	CODA
talent/documents/Army_to_College_Basketball_Replication_Handoff.md	CODA → SCOUT replication design
tenure/documents/TENURE_DATA_GAMEPLAN.md	PEER
PEER_Status_Update_for_VECTOR_2026-06-03.md	PEER

Stale / low-priority for Master Planner (unless scope expands)

- sports/documents/sports_mechanisms/advisor_packet*.md — older replication packets; may predate 538D.
- sports/documents/<!-- Generated by SpecStory, Markdown v2.md — chat export artifact; not a contract.
- Multiple alex_gates_briefing_structure_outline_v[1-3].md — superseded by v4 for oral flow.

6. Sister agents — what SCOUT knows (read their reports; don't merge pipelines)

Agent	Report	Maturity vs SCOUT	SCOUT dependency
CODA	<code>CODA_report_to_master_planner.md</code>	Most mature empirical + survival stack	Inverted-U origin ; CIF/competing-risks vocabulary
PEER	<code>PEER_report_to_master_planner.md</code>	Preliminary third setting (stage 9 bins)	Parallel LOO pool logic; tenure vs attrition
VECTOR	<i>(pending)</i>	Manuscript / theory	Consumes SCOUT status docs; does not run notebooks

SCOUT does not: merge Army OER snapshots into basketball panel, or tenure Wayback HTML into 530, or write dissertation chapter prose (VECTOR's lane).

7. Harmonization notes for Master Planner (language only — not code merge)

Concept	Army (CODA)	Basketball (SCOUT)	Academia (PEER)
Pool	Senior-rater cohort	Teammates (team-season)	Dept assistant cohort
LOO peer quality	Pool mean minus self	<code>poolq_loo</code>	<code>poolq_loo_mean</code>
Outcome	Promotion / separation	<code>Y_draft</code>	<code>tenure_event</code> / <code>attrition</code>
Inverted-U evidence	Established CIF panels	Established ventiles	Preliminary 18-bin stage 9
Generative layer	Less central in CODA report	Active (538D CELL 10)	Not SCOUT's scope
Competing risks framing	Native (Cell 11)	Draft yes/no (no time-to-event in ventile EDA)	Cox wired (Cells 10/10.5)

Shared mechanism phrase (VECTOR/Dakota): "Advancement under constrained distinction" — developmental benefits vs competitive constraints; finite distinction capacity in local comparison environments.

Wang ordering (cross-domain): (1) stylized fact, (2) minimal mechanism, (3) non-obvious predictions. SCOUT is between (2) and (3) on basketball generative axis.

8. Prior plan-map work (Cursor artifact — not master plan)

Charles and SCOUT previously mapped the doc ecosystem in `.cursor/plans/tier1_plan_map_c0d22e58.plan.md`. Key conclusion: **no single master plan file existed** as of June 2026; Dakota v03 RTF is the best **committee-facing whole-project narrative**; `Alex_Tier1_Sequential_Model_Outline.md` is the best **modeling spine** for basketball Tier 1.

Master Planner may absorb that map or supersede it — SCOUT does not treat the `.cursor/plans/` file as canonical repo documentation.

9. Clarifying questions for Charles (before Master Planner locks scope)

SCOUT would rather block on answers than guess. Grouped by theme.

A. Master Planner process

1. Will **one** Cursor agent ingest **all four** reports (CODA, SCOUT, PEER, VECTOR) and produce a single `MASTER_PLAN.md`, or separate plans per domain merged later?
2. Should the master plan target the **full dissertation** (three settings + generative + network extensions) or **Paper 1** scope (e.g. cross-domain empirical + minimal generative only)?
3. Is there a **deadline** (committee meeting, submission, grant) that should drive sequencing?
4. Should VECTOR's handoff be a **fifth markdown file** in `3-Master_Plan/`, or will Master Planner read `5-Manuscript/` directly?

B. Basketball generative priorities

5. What is the **next scientific priority** for SCOUT: (a) close L_Q generative gap, (b) CELL 4D heterogeneity narrative, (c) 538 empirical CELL 7 robustness, (d) HPC sweep over (w, τ, θ) , or (e) manuscript support only?
6. Should Master Planner treat **539** as deprecated, reference-only, or still active for Alex demos?
7. Is **team_mean** vs L_Q axis choice a **manuscript figure requirement** (show both) or internal lab detail?

C. Empirical basketball

8. Is `535` still an active notebook path, or is all Tier 1 empirical work consolidating in **538/538D**?
9. **PERF_METRIC lock**: Is PPM+z within-season still the canonical spec for cross-domain comparability, or should BPM/OBPM become primary before master plan freezes?

D. Cross-agent coordination

10. Should PEER adopt SCOUT's **generative modular stack** for academia simulation (future), or keep PEER empirical-only until tenure Cox results land?
11. Does CODA's **competing-risks** framing need a basketball analog (e.g. draft vs undrafted vs early exit) in the master plan, or is ventile draft-rate sufficient for Setting 2?

E. VECTOR / manuscript

12. Who **owns** updating Dakota RTF when PEER/CODA/SCOUT status shifts — VECTOR, Charles, or implementation agents?
13. Should the master plan index `advancement_under_constrained_distinction_dakota_feedback_v03.rtf` as the external-facing spine, with internal gameplans as appendices?

10. Suggested first tasks for Master Planner (after assignment only — SCOUT does not authorize these yet)

1. Read four agent reports + Dakota RTF + `Alex_Tier1_Sequential_Model_Outline.md`.
2. Produce a **one-page index** doc in `3-Master_Plan/` linking: gameplans, status updates, primary notebooks, canonical outcomes per setting.
3. Lock **claim language** table (what each setting can say in manuscript v1).
4. Sequence **open scientific work** (generative L_Q gap vs PEER Cox vs CODA TB-stratify) against Charles's timeline answer to §9A.3.
5. Assign **single owner** per cross-cutting doc (who refreshes Dakota, Alex spine, gameplans).
6. **Do not** merge `talent_pipeline/`, `sports_pipeline/`, and `tenure_pipeline/` codebases — harmonize at **estimand + narrative** layer only unless Charles explicitly requests shared Python abstractions.

11. Agent identity note

SCOUT in Charles's naming = Cursor agent focused on `./sports/`, college basketball data, Tier 1 generative modules, and SCOUT-authored status docs in `5-Manuscript/`. SCOUT implements, documents, and stress-tests; **VECTOR** drafts manuscript prose; **Alex Gates** sets sequencing; **Charles** owns scientific claims and committee communication.

If a future Cursor session is labeled "SCOUT," it should read this report + `Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md` before changing generative defaults or briefing language.

SCOUT handoff report ends. Master Planner: do not plan until Charles assigns you. VECTOR report still outstanding in `3-Master_Plan/` as of 2026-06-08.